Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Arrange the following in increasing order of ionic radii? $C^{4-}, N^{3-}, F^{-}, O^{2-}$

(A) $C^{4-} < N^{3-} < O^{2-} < F^{-}$

(B) $N^{3-} < C^{4-} < O^{2-} < F^{-}$

(C) $F^{-} < O^{2-} < N^{3-} < C^{4-}$

(D) $O^{2-} < F^{-} < N^{3-} < C^{4-}$

Q.2

In which of the following arrangements, the order is NOT according to the property indicated against it?

(A) $Li < Na < K < Rb$:
Increasing metallic radius

(B) $I < Br < F < Cl$:
Increasing electron gain enthalpy (with negative sign)

(C) $B < C < N < O$
Increasing first ionization enthalpy

(D) $Al^{3+} < Mg^{2+} < Na^{+} < F^{-}$
Increasing ionic size

Q.3

The incorrect statement among the following is

(A) The first ionization potential of Al is less than the first ionization potential of Mg

(B) The second ionization potential of Mg is greater than the second ionization potential of Na

(C) The first ionization potential of Na is less than the first ionization potential of Mg

(D) The third ionization potential of Mg is greater than the third ionization potential of Al.

Q.4

Following statements regarding the periodic trends of chemical reactivity of the alkali metals and the halogens are given. Which of these statements gives the correct picture?

(A) Chemical reactivity increases with increase in atomic number down the group in both the alkali metals and halogens

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(B) In alkali metals the reactivity increases but in the halogens it decreases with increase in atomic number down the group

(C) The reactivity decreases in the alkali metals but increases in the halogens with increase in atomic number down the group

(D) In both the alkali metals and the halogens the chemical reactivity decreases with increase in atomic number down the group

Q.5

Which one of the following arrangements represents the correct order of least negative to most negative electron gain enthalpy for $\mathrm{C}, \mathrm{Ca}, \mathrm{Al}, \mathrm{F}$ and O?

(A) $\mathrm{Ca} < \mathrm{Al} < \mathrm{C} < \mathrm{O} < \mathrm{F}$

(B) $\mathrm{Al} < \mathrm{Ca} < \mathrm{O} < \mathrm{C} < \mathrm{F}$

(C) $\mathrm{Al} < \mathrm{O} < \mathrm{C} < \mathrm{Ca} < \mathrm{F}$

(D) $\mathrm{C} < \mathrm{F} < \mathrm{O} < \mathrm{Al} < \mathrm{Ca}$

Q.6

Similarity in chemical properties of the atoms of elements in a group of the Periodic table is most closely related to:

(A) atomic numbers

(B) atomic masses

(C) number of principal energy levels

(D) number of valence electrons

Q.7

Sum of first three ionization energies of Al is $53.0\mathrm{eV}$ atom $^{-1}$ and the sum of first two ionization energies of Na is $52.2\mathrm{eV}$ atom $^{-1}$. Out of Al(III) and $\mathrm{Na(II)}$

(A) Na (II) is more stable than Al (III)

(B) $\mathrm{Al(III)}$ is more stable than $\mathrm{Na(II)}$

(C) Both are equally stable

(D) Both are equally unstable

Q.8

The correct order of decreasing electronegatıvity values among the elements I-beryllium, II-oxygen, III-nitrogen and IV-magnesium is

(A) $\mathrm{II} > \mathrm{III} > \mathrm{I} > \mathrm{IV}$

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(B) $\mathrm{III} > \mathrm{IV} > \mathrm{II} > \mathrm{I}$

(C) $\mathrm{I} > \mathrm{II} > \mathrm{III} > \mathrm{IV}$

(D) $\mathrm{I} > \mathrm{II} > \mathrm{IV} > \mathrm{III}$

Q.9

First three ionisation energies (in $\mathrm{kJ/mol}$) of three representative elements are given below:

Element	IE_1	IE_2	IE_3
P	495.8	4562	6910
Q	737.7	1451	7733
R	577.5	1817	2745

Then incorrect option is :

(A) Q : Alkaline earth metal

(B) P : Alkali metals

(C) R : s -block element

(D) They belong to same period

Q.10

Which of the following sequence correctly represents the decreasing acidic nature of oxides?

(A) $\mathrm{Li_2O} > \mathrm{BeO} > \mathrm{B_2O_3} > \mathrm{CO_2} > \mathrm{N_2O_3}$

(B) $\mathrm{N_2O_3} > \mathrm{CO_2} > \mathrm{B_2O_3} > \mathrm{BeO} > \mathrm{Li_2O}$

(C) $\mathrm{CO_2} > \mathrm{N_2O_3} > \mathrm{B_2O_3} > \mathrm{BeO} > \mathrm{Li_2O}$

(D) $\mathrm{B_2O_3} > \mathrm{CO_2} > \mathrm{N_2O_3} > \mathrm{Li_2O} > \mathrm{BeO}$

Q.11

Correct order of first IP among following elements $\mathrm{Be, B, C, N, O}$ is

(A) $\mathrm{B} < \mathrm{Be} < \mathrm{C} < \mathrm{O} < \mathrm{N}$

(B) $\mathrm{B} < \mathrm{Be} < \mathrm{C} < \mathrm{N} < \mathrm{O}$

(C) $\mathrm{Be} < \mathrm{B} < \mathrm{C} < \mathrm{N} < \mathrm{O}$

(D) $\mathrm{Be} < \mathrm{B} < \mathrm{C} < \mathrm{O} < \mathrm{N}$

Q.12

Aqueous solutions of two compounds $\mathrm{M_1 - O - H}$ and $\mathrm{M_2 - O - H}$ are prepared in two different beakers. If, the electronegativity of $\mathrm{M_1} = 3.4, \mathrm{M_2} = 1.2, \mathrm{O} = 3.5$ and $\mathrm{H} = 2.1$ then the nature of two solutions will be respectively :

(A) acidic, basic

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(B) acidic, acidic

(C) basic, acidic

(D) basic, basic

Q.13

An element having electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$ forms

(A) Acidic oxide

(B) Basic oxide

(C) Amphoteric oxide

(D) Neutral oxide

Q.14

Which among the following elements has the highest first ionization enthalpy?

(A) Nitrogen

(B) Boron

(C) Carbon

(D) Oxygen

Q.15

Which one of the following statements is incorrect in relation to ionisation enthalpy?

(A) Ionisation enthalpy increases for each successive electron.

(B) The greatest increase in ionisation enthalpy is experienced on removal of electron from core noble gas configuration.

(C) End of valence electrons is marked by a big jump in ionisation enthalpy.

(D) Removal of electron from orbitals bearing lower n value is easier than from orbital having higher n value.

Q.16

If the ionization enthalpy and electron gain enthalpy of an element are 275 and $86 \mathrm{kcal mol}^{-1}$ respectively, then the electronegativity of the elementon the Pauling scale is:

(A) 2.8

(B) 0.0

(C) 4.0

(D) 2.6

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Q.17

If the difference in electronegativities of two elements is very large, then

(A) The bond is 50% ionic

(B) The bond is 100% covalent

(C) The bond is more covalent than ionic

(D) The bond is more ionic than covalent

Q.18

In which of the following arrangements the order is NOT according to the property indicated against it

(A) $\mathrm{Al}^{3+} < \mathrm{Mg}^{2+} < \mathrm{Na}^{+} < \mathrm{F}^{-}$: Increasing ionic size

(B) $\mathrm{B} < \mathrm{C} < \mathrm{N} < 0$: Increasing first ionization enthalpy

(C) $I < Br < F < Cl$: Increasing electron gain enthalpy (with negative sign)

(D) $\mathrm{Li} < \mathrm{Na} < \mathbf{K} < \mathrm{Rb}$ **:** Increasing metallic radius

Q.19

Under normal condition which of the following electronic configuration is able to form dipositive ion

(A) $[\mathrm{Ar}] 4\ \mathrm{s}^1$

(B) $[\mathrm{Ne}] 2\ \mathrm{s}^2 3\mathrm{p}^6$

(C) $[\mathrm{Ne}] 3\ \mathrm{s}^2$

(D) None of these

Q.20

Which of the following orders is correct for the size?

(1) $\mathrm{Mg}^{2+} < \mathrm{Na}^{+} < \mathrm{F}^{-} < \mathrm{Al}$

(2) $\mathrm{Te}^{2-} > \mathrm{I} > \mathrm{Cs}^{+} > \mathrm{Ba}^{2+}$

(3) $\mathrm{Fe}^{3+} < \mathrm{Fe}^{2+} < \mathrm{Fe}^{4+}$

(4) $\mathrm{Mg} > \mathrm{Al} > \mathrm{Si} > \mathrm{P}$

(A) $(1), (2) \& (4)$

(B) $(2), (3) \& (4)$

(C) (1), (3) \& (4)

(D) $(1), (2), (3) \& (4)$

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Q.1 Answer:

$F^- < O^{2-} < N^{3-} < C^{4-}$

Solution:

All the given species are isoelectronic. In case of isoelectronic species ionic radii increases with increase in negative charge on anions.

Q.2 Answer:

$B < C < N < O$

Increasing first ionization enthalpy

Solution:

In a period the value of ionisation potential increases from left to right with breaks where the atoms have some what stable configuration. In this case N has half filled stable orbitals. Hence has highest ionisation energy. Thus the correct order is

$B < C < O < N$

and not as given in option (c)

Q.3 Answer:

The second ionization potential of Mg is greater than the second ionization potential of Na

Solution:

IE_2 of Mg is lower than that of Na because in case of Mg^+, $3s$ -electron has to be removed whereas in case of Na^+, an electron is removed from the stable inert gas configuration which is difficult.

Q.4 Answer:

In alkali metals the reactivity increases but in the halogens it decreases with increase in atomic number down the group

Solution:

The alkali metals are highly reactive because their first ionisation potential is very low and hence they have great tendency to loses electron to form unipositive ion. On moving down group- I from Li to Cs ionisation enthalpy decreases hence the reactivity increases. The halogens are most reactive elements due to their low bond dissociation energy, high electron affinity and high enthalpy of hydration of halide ion. However their reactivity decreases with increase in atomic number.

Q.5 Answer:

$Ca < Al < C < O < F$

Solution:

As the nuclear charge increases, the force of attraction between the nucleus and the incoming electron increases and hence the elecron gain enthalpy becomes more negative, hence the correct order is $Ca < Al < C < O < F$

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Q.6 Answer:

number of valence electrons

Solution:

The chemical properties of the atoms of elements in a group of the periodic table is most closely related to the number of valence electrons. All the chemical properties can be justified just by the number of valence electrons.

Q.7 Answer:

$\mathrm{Al(III)}$ is more stable than $\mathrm{Na(II)}$

Solution:

Ionization energy is not the only the criteria for the stability of an oxidation state.

Q.8 Answer:

$\mathrm{II} > \mathrm{III} > \mathrm{I} > \mathrm{IV}$

Solution:

Electronegativity values of given elements are as follows:

$\mathrm{Be} - 1.5(\mathrm{I}) \quad \mathrm{Mg} - 1.2(\mathrm{IV})$

$\mathrm{O} - 3.5\ (\mathrm{II}) \quad \mathrm{N} - 3.0\ (\mathrm{III})$

i.e. $\mathrm{II} > \mathrm{III} > \mathrm{I} > \mathrm{IV}$

Q.9 Answer:

R : s -block element

Solution:

R is p -block element, because difference between IE_2 and IE_3 is not very high as compared to between IE_1 and IE_2; hence stable oxidation state of R will be higher than $+2$.

Q.10 Answer:

$\mathrm{N_2O_3} > \mathrm{CO_2} > \mathrm{B_2O_3} > \mathrm{BeO} > \mathrm{Li_2O}$

Solution:

On passing from left to right in a period, acidic character of the normal oxides of the elements increases with increase in electronegativity.

Q.11 Answer:

$\mathrm{B} < \mathrm{Be} < \mathrm{C} < \mathrm{O} < \mathrm{N}$

Solution:

$\mathrm{Be} - 1s^2 2s^2; \mathrm{B} - 1s^2 2s^2 2p^1; \mathrm{C} - 1s^2 2s^2 2p^2$

$\mathrm{N} - 1s^2 2s^2 2p^3; \mathrm{O} - 1s^2 2s^2 2p^4$**. IP increases**

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along the period. But IP of $\mathrm{Be} > \mathrm{B}$. Further IP of $\mathrm{O} < \mathrm{N}$ because atoms with fully or partly filled orbitals are most stable and hence have high ionisation energy.

Q.12 Answer:

acidic, basic

Solution:

The electronegativity difference between M_1 and O is 0.1, which indicates $\mathrm{M}_1 - \mathrm{O}$ bond will be covalent, since $\mathrm{O} - \mathrm{H}$ bond having more ionic character thus bond will break and H^+ ions gets release and acidic solution is formed and whereas difference between electronegativity of $\mathrm{M}_2\mathrm{O}$ is 2.3, thus, $\mathrm{M}_2 - \mathrm{OH}$ bond will break. Hence, solution will be basic in nature.

Q.13 Answer:

Basic oxide

Solution:

It is electronic configuration of alkali metal. Hence it will form basic oxide.

Q.14 Answer:

Nitrogen

Solution:

Due to stable $2s^2 2p^3$ configuration (half filled p -orbital). Nitrogen atom has highest energy.

Q.15 Answer:

Removal of electron from orbitals bearing lower n value is easier than from orbital having higher n value.

Solution:

As the value of n increases, the nuclear attraction over the outermost shell decreases, therefore removal of an electron will be easier.

Q.16 Answer:

2.8

Solution:

I.E. $+$ **E.A.** $= 275 + 86 = 361 \mathrm{kcal mol}^{-1}$

$= 361 \times 4.184 = 1510.42 \text{ kJ mol}^{-1}$

$\therefore \quad$ **Electronegativity** $= \frac{1510.42}{540}$

$= 2.797 = 2.8$

Q.17 Answer:

The bond is more ionic than covalent

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Solution:

If the difference in electronegativities of two elements is very high then the bond is more ionic than covalent.

Q.18 Answer:

$B < C < N < 0$: Increasing first ionization enthalpy

Solution:

$B < C < N < O$; **When we move from** B **to** O**in a periodic table the first ionization enthalpy increase due to the attraction of nuclens towards the outer most electron and IE of** $N > 0$**.**

Q.19 Answer:

$[Ne]3\ s^2$

Solution:

Mg **has only two electrons in the 3 s-orbital and hence its** $LE.$ **is lowest, i.e. it has the maximum tendency to from di-positive ions.**

Q.20 Answer:

$(1), (2) \& (4)$

Solution:

(1) Mg^{2+}, Na^{+} **and** F^{-} **are isoelectronic and thus follows the order** ${}_{12}Mg^{2+} <_{11} Na^{+} <_{q} F$

Al belongs to third period and has no charge so it is largest. $Na^{+} = 102pm, Mg^{2+} = 72pm,$ **Al** $= 143pm, F^{-} = 133pm$

(2) Isoelectronic series of ions, all have the Xenon electron configuration

Ionic radius $\propto \frac{1}{\text{muclear charge}}$

Atomic number $Te = 52, I = 53, Cs = 55, Ba = 56$

(4) Across the period the muclear charge increases and thus the size of atoms decreases.

$Mg = 160pm, Al = 143pm, Si = 118, \mathbf{P} = 110pm$

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